

CoRoT-1b

R-2638

Benchmark run · workflow W8-benchmark-deep · record deep-bench_CoRoT-1_130401 · created 2026-08-29T01:03:33+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2456383.6695 +/- 0.0021 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.064 +/- 0.04
Transit depth (Rp/Rs)^2	0.41 +/- 0.51 [%]
Orbital Inclination (inc)	88.58 +/- 2.83
Airmass coefficient 1 (a1)	1.136 +/- 0.031
Airmass coefficient 2 (a2)	-0.0382 +/- 0.0097
Scatter in the residuals of the lightcurve fit is	2.73 %
Best Comparison Star	#3 - [276, 82]
Optimal Aperture	2.59
Optimal Annulus	13.45
Transit Duration (day)	0.0946 +/- 0.0041

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.064 ± 0.04 vs 0.1388 (deviation 1.87σ)
Duration fit vs published	0.0946 d vs 0.1046 d (deviation 2.43σ)
Mid-time O–C	-5.0 min
Depth SNR	1.6
Residual scatter	2.73 %

Light curve

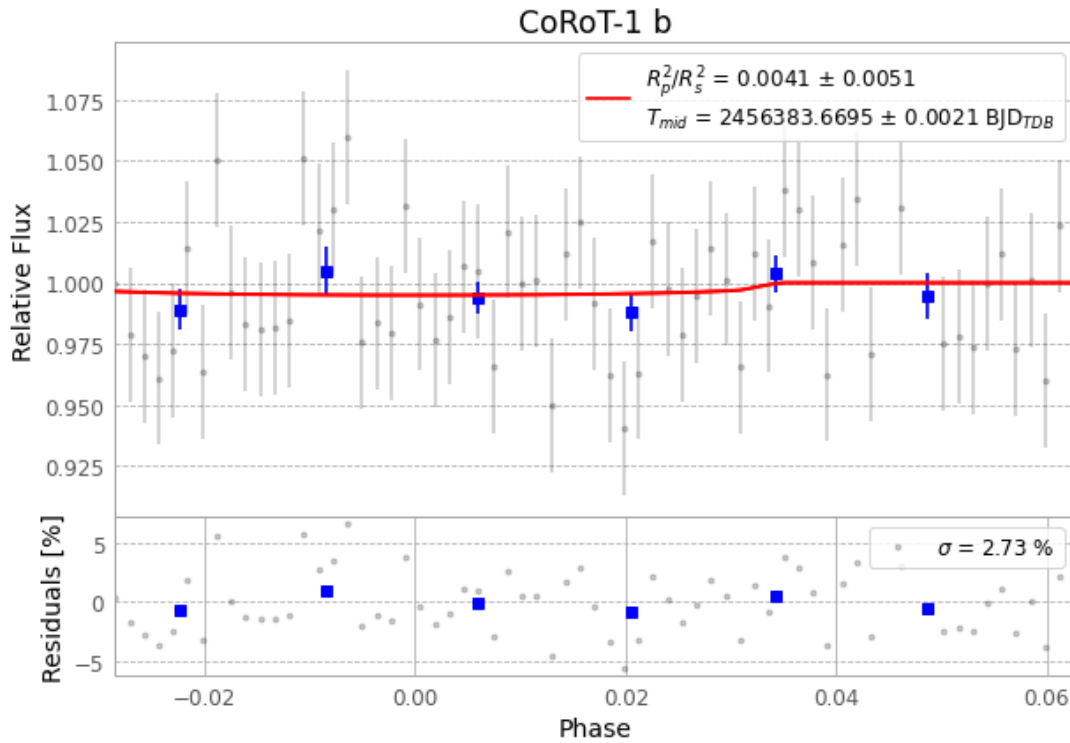


Figure 1 — FinalLightCurve_CoRoT-1 b_2013-03-31.png (SHA-256 92feb7d152c06263...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [330, 309], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 14ba936b8f5dba55...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit a4253ce

Data provenance

67 FITS frames (44 MB), CoRoT-1130401030012.FITS → CoRoT-1130401061812.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-CoRoT-1-130401.log (a5df91b8488e...), FinalLightCurve_CoRoT-1 b_2013-03-31.png (92feb7d152c0...), FinalLightCurve_CoRoT-1 b_2013-03-31.csv (9b7e53cfd926...)

Compute resources

Wall time 112.9 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-29 01:03Z.